

V6.2 · Soft Theme Membership

Every motif carries a distribution over chemical themes, not a label

Objective	Replace hard motif→theme assignment with a non-negative, row-stochastic, sparse membership matrix S .
Method	$S[m, \cdot]$ = softmax over the cosine between motif m 's corpus activation profile and each theme centroid, followed by a floor at 0.02 and renormalisation.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cde; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Method

p_m = motif m 's activation across the 167 analytes, L2-normalised
 c_t = mean p_m over theme t 's seed motifs, L2-normalised
 $S[m, t] \propto \exp(\cos(p_m, c_t) / \tau)$, then floor at 0.02 and renormalise

τ is chosen by a criterion fixed in advance: the largest (softest) temperature on a fixed grid for which the mean number of themes carrying a motif is ≤ 2.5 — "a motif belongs to one, sometimes two chemistries". The full sweep is reported so the choice is auditable.

τ	mean support	mean entropy	mean dominant weight	
0.02	1.18	0.028	0.984	
0.03	1.29	0.053	0.966	
0.04	1.76	0.114	0.932	
0.05	2.06	0.166	0.898	◀ selected
0.07	2.77	0.280	0.824	
0.1	4.18	0.468	0.707	
0.12	4.94	0.574	0.640	
0.15	5.82	0.710	0.551	
0.2	6.00	0.827	0.453	

Parts 1-2 — soft membership and theme uncertainty

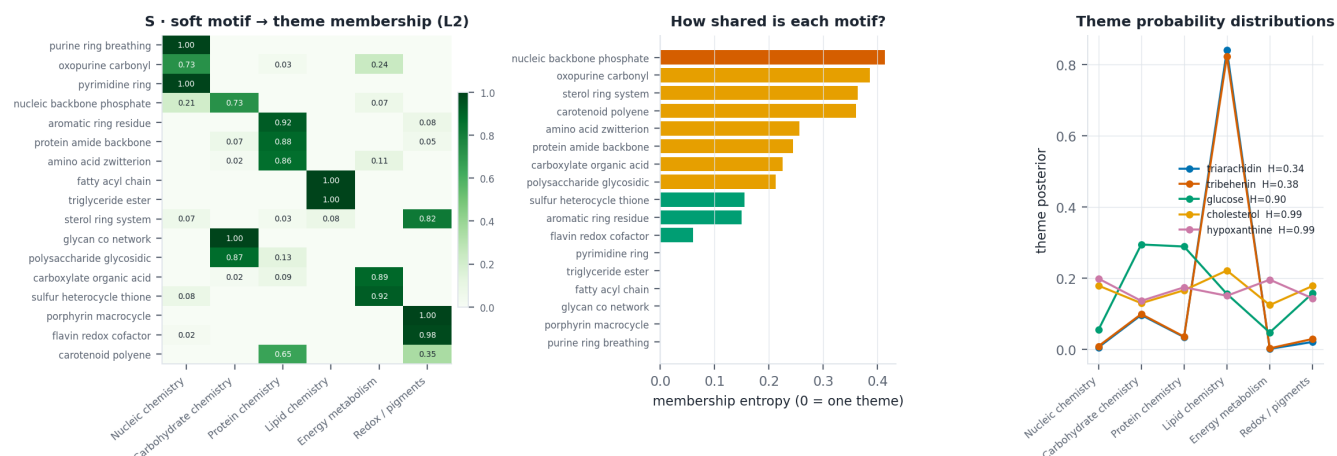


Figure 1 — The membership matrix, per-motif entropy, and example theme posteriors.

Results

motif	dominant theme	w	runner-up	w	entropy	support
purine ring breathing	Nucleic chemistry	1.000	Carbohydrate chemistry	0.000	-0.000	1
oxopurine carbonyl	Nucleic chemistry	0.725	Energy metabolism	0.240	0.386	3
pyrimidine ring	Nucleic chemistry	1.000	Carbohydrate chemistry	0.000	-0.000	1
nucleic backbone phosphate	Carbohydrate chemistry	0.726	Nucleic chemistry	0.207	0.413	3
aromatic ring residue	Protein chemistry	0.924	Redox / pigments	0.076	0.150	2
protein amide backbone	Protein chemistry	0.883	Carbohydrate chemistry	0.071	0.245	3
amino acid zwitterion	Protein chemistry	0.865	Energy metabolism	0.111	0.256	3
fatty acyl chain	Lipid chemistry	1.000	Nucleic chemistry	0.000	-0.000	1
triglyceride ester	Lipid chemistry	1.000	Nucleic chemistry	0.000	-0.000	1
sterol ring system	Redox / pigments	0.823	Lipid chemistry	0.079	0.363	4
glycan co network	Carbohydrate chemistry	1.000	Nucleic chemistry	0.000	-0.000	1
polysaccharide glycosidic	Carbohydrate chemistry	0.873	Protein chemistry	0.127	0.212	2
carboxylate organic acid	Energy metabolism	0.890	Protein chemistry	0.085	0.225	3
sulfur heterocycle thione	Energy metabolism	0.921	Nucleic chemistry	0.079	0.155	2
porphyrin macrocycle	Redox / pigments	1.000	Nucleic chemistry	0.000	-0.000	1
flavin redox cofactor	Redox / pigments	0.977	Nucleic chemistry	0.023	0.060	2
carotenoid polyene	Protein chemistry	0.654	Redox / pigments	0.346	0.360	2

Two motifs land on the wrong dominant theme, and the soft layer makes it visible rather than hiding it. *sterol_ring_system* is dominated by Redox / pigments (0.82) rather than Lipid chemistry (0.08), and *carotenoid_polyene* by Protein chemistry (0.65) over Redox / pigments (0.35). Both are corpus-limited motifs already flagged in V6; the membership entropy now quantifies the problem instead of a hard label concealing it.

Interpretation

Mean membership entropy at L2 is 0.166 and the mean dominant weight 0.898. 6 motifs are cleanly single-theme (purine, pyrimidine, monosaccharide, fatty acyl, acylglycerol, porphyrin) — these are the chemistries the corpus grounds well. The rest carry real, interpretable overlap.

Limitations

- τ is a free parameter; a different sparsity target would give a different S.
- The theme centroids are computed from the seed grouping, so S inherits any error in that grouping — most visibly for sterol and carotenoid.
- Membership is derived from corpus co-activation, so it describes THIS corpus.

Recommendations

- Report the entropy alongside every theme call; a high-entropy motif should not drive an interpretation on its own.
- Re-derive S after adding pure sterol/porphyrin/flavin references.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.